DESCENT DATA FOR QUASI-COHERENT SHEAVES OVER AN OPEN COVERING ARE EFFECTIVE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted spans carry navigation stubs for the references leaving them; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- Glueing of sheaves along an open covering. Sheaves, Section sheaves-section-glueing-sheaves, invoked at proof lines 404-405. Produces the glued sheaf and the fully faithful comparison functor.
- Quasi-coherence of modules is a local property. Modules, Definition modules-definition-quasi-coherent, invoked at proof lines 406-407. Ensures the glued sheaf is quasi-coherent, so the descent datum is effective in the required category.

Source reference. Original definition 01BE: sources/stacks-chow/tags.txt, source lines 1723–1723. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original section 00AK: sources/stacks-chow/tags.txt, source lines 392–392. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let S be a scheme. Let $S = \bigcup U_i$ be an open covering. Any descent datum on quasi-coherent sheaves for the family $\mathcal{U} = \{U_i \to S\}$ is effective. Moreover, the functor from the category of quasi-coherent $\mathcal{O}_S$-modules to the category of descent data with respect to $\mathcal{U}$ is fully faithful.*

Proof. This follows immediately from Sheaves, Section 00AK and the fact that being quasi-coherent is a local property, see Modules, Definition 01BE. □

Date: stacks-project:descent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.